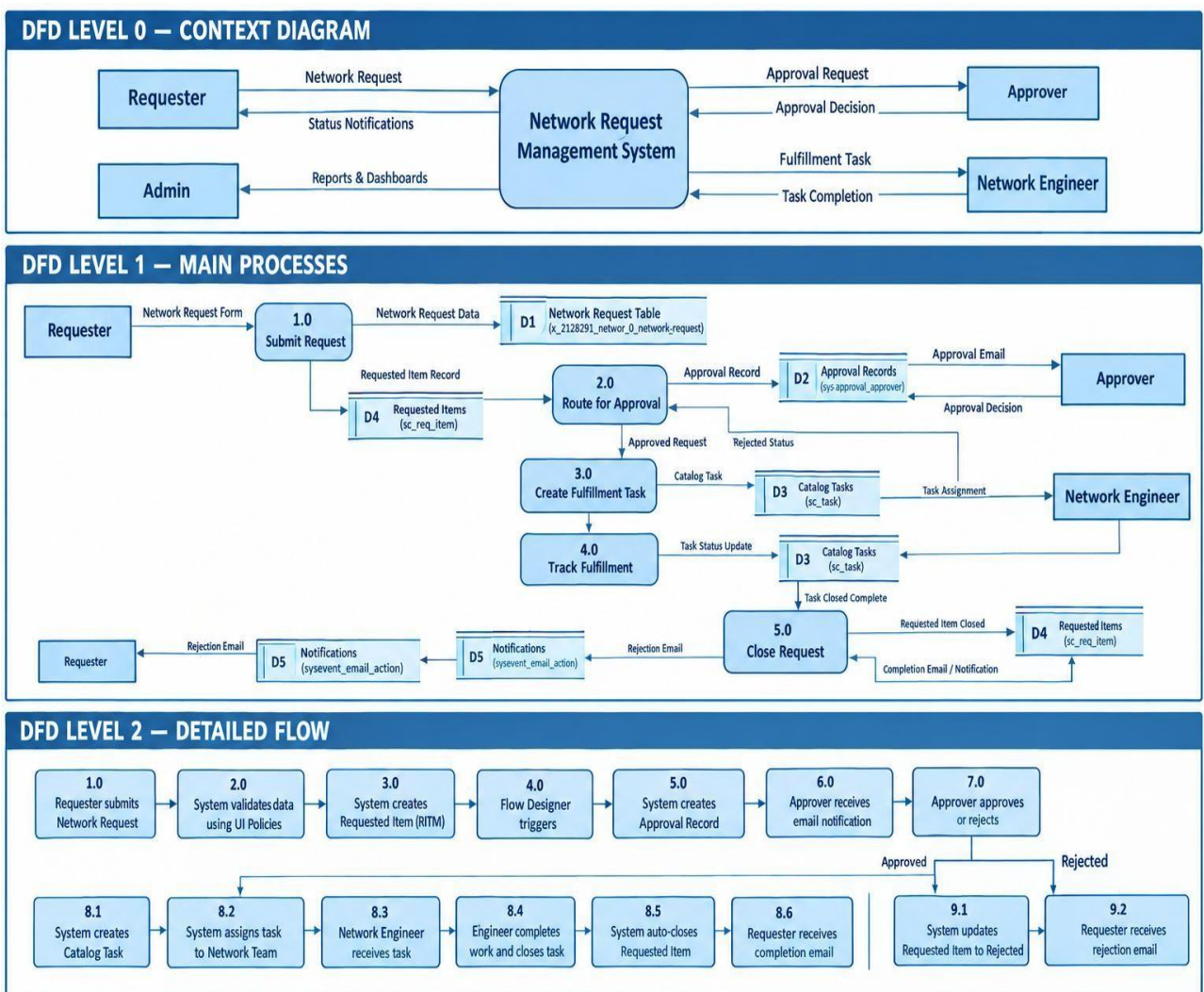


Date	14 September 2026
Team ID/SID	SWUID2026136950
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:



This Data Flow Diagram (DFD) represents the complete flow of data in the Automated Network Request Management System. It shows how a network request moves through the system — from submission by the requester to approval, task creation, fulfillment, and final

closure. The diagram is split into three levels: Level 0 (Context Diagram), Level 1 (Main Processes), and Level 2 (Detailed Flow).

DFD Level 0 — Context Diagram

The Level 0 DFD, also known as the Context Diagram, shows the system as a single central process surrounded by four external entities: Requester, Approver, Network Engineer, and Admin. The Requester sends a Network Request into the system and receives Status Notifications back. The Approver receives an Approval Request and sends back an Approval Decision. The Network Engineer receives a Fulfillment Task and sends back Task Completion. The Admin receives Reports and Dashboards from the system. This high-level view defines the system boundary and the interactions with external users.

DFD Level 1 — Main Processes

The Level 1 DFD breaks the system into five main processes: Submit Request, Route for Approval, Create Fulfillment Task, Track Fulfillment, and Close Request. It also introduces five data stores: D1 (Network Request Table), D2 (Approval Records), D3 (Catalog Tasks), D4 (Requested Items), and D5 (Notifications).

Process 1.0 (Submit Request) takes the Network Request Form from the Requester and creates a Requested Item in D4. The data is also stored in D1.

Process 2.0 (Route for Approval) takes the Requested Item and creates an Approval Record in D2. An Approval Email is sent to the Approver.

Process 3.0 (Create Fulfillment Task) takes the approved request and creates a Catalog Task in D3. The task is assigned to the Network Engineer.

Process 4.0 (Track Fulfillment) captures the work notes and status updates from the Network Engineer and updates D3.

Process 5.0 (Close Request) takes the completed task and closes the Requested Item in D4. A completion or rejection email is sent through D5 to the Requester.

DFD Level 2 — Detailed Flow

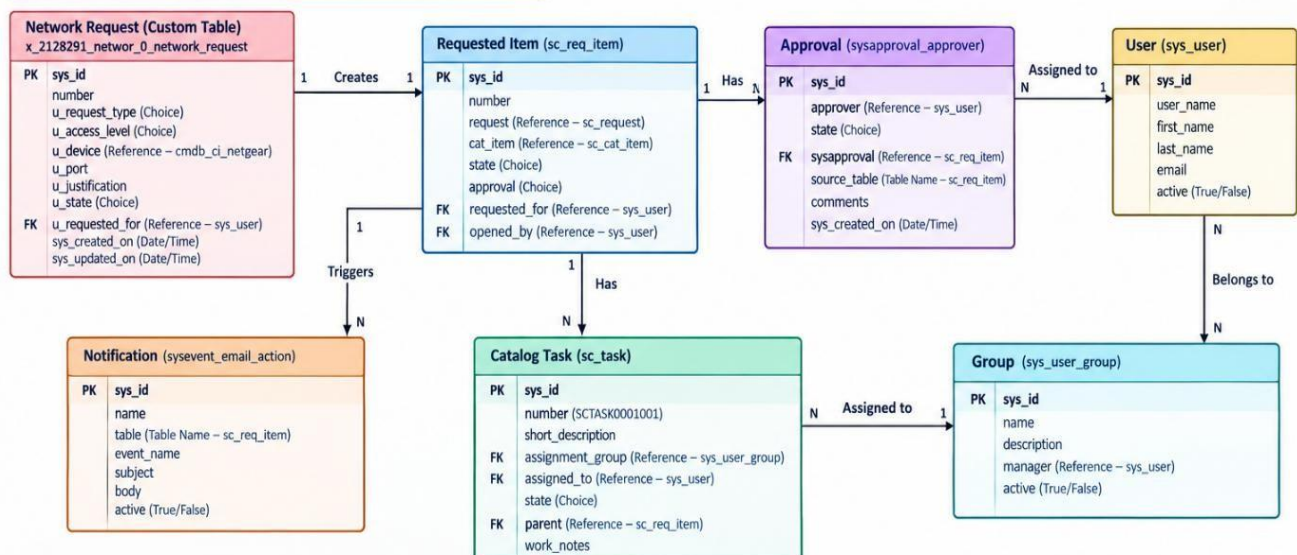
The Level 2 DFD provides a step-by-step view of the entire workflow. It starts with the Requester submitting a Network Request. The system validates the data using UI Policies and

creates a Requested Item (RITM). The Flow Designer then triggers automatically, creating an Approval Record and sending an email notification to the Approver. The Approver reviews the request and either approves or rejects it.

If approved, the system creates a Catalog Task, assigns it to the Network Team, and the Network Engineer receives it. Once the engineer completes the work and closes the task, the system auto-closes the Requested Item and sends a completion email to the Requester.

If rejected, the system updates the Requested Item to Rejected and sends a rejection email to the Requester.

ER DIAGRAM — ENTITY RELATIONSHIP



Relationship Summary

Entity 1	Relationship	Entity 2	Cardinality
Network Request	Creates	Requested Item	1 : 1
Requested Item	Has	Approval	1 : N
Requested Item	Has	Catalog Task	1 : N
Approval	Assigned to	User	N : 1
Catalog Task	Assigned to	Group	N : 1
Catalog Task	Assigned to	User	N : 1
Requested Item	Triggers	Notification	1 : N
User	Belongs to	Group	N : N

Legend

PK	= Primary Key	→	= Relationship
FK	= Foreign Key	1	= One (Cardinality)
		N	= Many (Cardinality)

Tables (ServiceNow)

Network Request (Custom Table)	Catalog Task (sc_task)
Requested Item (sc_req_item)	User (sys_user)
Approval (sysapproval_approver)	Group (sys_user_group)
	Notification (sysevent_email_action)

User story:

User Type	Functional Requirement(Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Requester	Network Request Submission	USN-1	As a requester, I can submit a network request through the Service Catalog so that I don't have to send emails.	I can fill the form, submit it, and receive a request number	High	Sprint-1
Requester	Dynamic Form Behavior	USN-2	As a requester, I can see only the relevant fields based on my selected request type.	Device and Port fields appear only when I select Firewall	High	Sprint-1
Requester	Mandatory Field Validation	USN-3	As a requester, I cannot submit the form without filling all required fields.	Form shows validation errors for empty mandatory fields	High	Sprint-1
Requester	Request Tracking	USN-4	As a requester, I can track the status of my request in the Service Portal.	I can see the current state (New, In Progress, Approved, Completed)	High	Sprint-2
Requester	Email Notifications	USN-5	As a requester, I receive email notifications at each stage of my request.	I receive emails when submitted, approved/rejected, and completed	High	Sprint-1
Requester	Auto-Closure	USN-6	As a requester, my request closes automatically when the task is completed.	The Requested Item state changes to Closed Complete	High	Sprint-1

Project Design Phase-II
Data Flow Diagram & User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Approver	Approval Workflow	USN-7	As an approver, I can approve or reject network requests assigned to me.	I can view the approval, add comments, and approve/reject	High	Sprint-1
Approver	Approval Notifications	USN-8	As an approver, I receive an email when a request needs my approval.	I receive an email with a direct link to approve or reject	High	Sprint-1
Approver	Multi-Level Approval	USN-9	As an approver, I can escalate high-risk requests to a second-level approver.	Firewall requests require two levels of approval	Medium	Sprint-3
Network Engineer	Task Assignments	USN-10	As a network engineer, I receive a Catalog Task when a request is approved.	The task is auto-created and assigned to my team	High	Sprint-1
Network Engineer	Task Fulfillment	USN-11	As a network engineer, I can complete the task and add work notes.	I can update the state to Closed Complete with work notes	High	Sprint-1
Network Engineer	Task Notifications	USN-12	As a network engineer, I receive an email when a task is assigned to me.	I receive an email with task details and a direct link	Medium	Sprint-2
Network Engineer	Work Note History	USN-13	As a network engineer, I can see all previous work notes on a task.	Activity stream shows all	Medium	Sprint-2

Project Design Phase-II
Data Flow Diagram & User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
				comments and work notes		
Admin	Access Control	USN-14	As an admin, I can control who can submit, approve, and fulfill requests.	Roles and ACLs are enforced at the record level	High	Sprint-1
Admin	Reporting	USN-15	As an admin, I can view reports on request volume and task performance.	I can see a bar chart of requests by state	Medium	Sprint-3
Admin	Dashboard	USN-16	As an admin, I can view a dashboard with key metrics.	I can see SLA compliance, task performance, and request volume	Medium	Sprint-3
Admin	Audit Trail	USN-17	As an admin, I can view the full audit trail of any request.	I can see who approved, when, and what changes were made	High	Sprint-2
Requester	Request Cancellation	USN-18	As a requester, I can cancel my request before approval.	I can click Cancel and the request moves to Cancelled state	Low	Sprint-4

Project Design Phase-II
Data Flow Diagram & User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Requester	Feedback	USN-19	As a requester, I can provide feedback after my request is completed.	I receive a survey link in the completion email	Low	Sprint-4
Requester	Request Again	USN-20	As a requester, I can re-submit a previous request with one click.	I can click "Request Again" and the form pre-fills with past data	Low	Sprint-4

Project Design Phase-II
Data Flow Diagram & User Stories